

Introduction to Data Science 11372 & 11516

Assignment 1 - Data Wrangling: Solution

Ruhan Shafi - u3284342

Your student ID (as recorded in Canvas): u3284342

Setup: Generating the dataset

Run this chunk exactly as provided. **Do not edit the generator.** If you change it, the dataset your marker regenerates will not match yours and your fingerprint check will fail.

```
knitr::opts_chunk$set(echo = FALSE)
## -----
## DO NOT EDIT THIS BLOCK
## -----

seed_from_id <- function(student_id) {
  id <- toupper(gsub("[^0-9A-Za-z]", "", as.character(student_id)))
  if (nchar(id) == 0L) stop("student_id must contain letters or digits.")
  ch <- utf8ToInt(id)
  h <- 5381 # djb2-style rolling hash
  for (k in ch) h <- (h * 33 + k) %% 2147483647
  as.integer(h %% 1000000000) + 1L
}

generate_weather_data <- function(student_id,
                                   start_date = as.Date("2024-01-01"),
                                   end_date   = as.Date("2025-12-31")) {

  RNGkind("Mersenne-Twister", "Inversion", "Rejection") # reproducible on R >= 3.6
  set.seed(seed_from_id(student_id))

  dates <- seq.Date(start_date, end_date, by = "day")
  n      <- length(dates)
  doy    <- as.numeric(format(dates, "%j"))
  yr     <- as.integer(format(dates, "%Y"))
  w      <- 2 * pi * (doy - 20) / 365.25 # temperature peak ~20 January

  ## 1. Rainfall: two-state Markov chain (wet and dry spells) -----
  seas_wet <- 1 + 0.18 * cos(2 * pi * (doy - 105) / 365.25) # autumn maximum
  p_dw    <- pmin(0.9, 0.260 * seas_wet) # P(wet today | dry yesterday)
  p_ww    <- pmin(0.9, 0.580 * seas_wet) # P(wet today | wet yesterday)
  wet     <- logical(n); wet[1] <- runif(1) < 0.3
  for (i in 2:n) wet[i] <- runif(1) < if (wet[i - 1]) p_ww[i] else p_dw[i]
```

```

rain <- ifelse(wet, rgamma(n, shape = 0.80, scale = 10.5), 0)
rain <- round(pmin(rain, 210), 1)
rain[rain < 0.2] <- 0 # below gauge resolution
wet <- rain > 0

## 2. Temperature: climatology + AR(1) anomaly + summer heatwave spikes ---
t_clim <- 18.8 + 5.3 * cos(w)
phi <- 0.74; sig <- 2.6 * sqrt(1 - phi^2)
anom <- numeric(n); anom[1] <- rnorm(1, 0, 2.6)
for (i in 2:n) anom[i] <- phi * anom[i - 1] + rnorm(1, 0, sig)

spike <- rbinom(n, 1, 0.012) * rexp(n, 1 / 3.2) * pmax(0, cos(w))^1.5
heat <- spike # smear over a 3-day heatwave
heat[-1] <- heat[-1] + 0.70 * spike[-n]
heat[-(1:2)] <- heat[-(1:2)] + 0.40 * spike[-((n-1):n)]
anom <- anom + pmin(heat, 7)

temp <- t_clim + anom - 1.6 * wet

## 3. Relative humidity -----
rh <- 65 + 2 * cos(w) - 2.1 * anom + 10 * wet + 6 * pmin(1, rain / 20) +
  rnorm(n, 0, ifelse(wet, 4.5, 6.5))
rh <- pmin(99, pmax(24, rh))

## 4. Wind speed -----
wind <- rweibull(n, shape = 2.1, scale = 15) + 3.5 * wet +
  1.2 * cos(2 * pi * (doy - 300) / 365.25)
wind <- pmax(1.5, wind)

## 5. Solar exposure: clear-sky cycle x cloud factor -----
ws <- 2 * pi * (doy - 172) / 365.25 # minimum at winter solstice
clear <- 22 - 10.0 * cos(ws)
cf <- ifelse(wet, rbeta(n, 2.4, 2.6), rbeta(n, 9, 1.4))
solar <- pmax(0.6, clear * cf)

## 6. Soil moisture: single-layer bucket with memory -----
sm <- numeric(n); sm[1] <- 30
for (i in 2:n) {
  et <- (0.12 + 0.035 * max(temp[i], 0)) * (solar[i] / 18) # evapotranspiration
  drain <- 0.032 * (sm[i - 1] - 8) # slow drainage
  sm[i] <- min(max(sm[i - 1] + 0.30 * rain[i] - et - drain, 8), 55)
}
sm <- pmin(56, pmax(5.5, sm + rnorm(n, 0, 0.35)))

## 7. Diurnal range and daily extremes -----
drange <- pmin(14, pmax(2.5, 6.0 + 3.2 * (solar / clear) - 0.03 * rh +
  0.40 * anom + rnorm(n, 0, 0.8)))
tmax <- temp + drange / 2

## 8. Fire danger: McArthur FFDI on afternoon conditions, binned 1-10 -----
drought <- 10 * (1 - (sm - 8) / (55 - 8))
rh_pm <- pmax(8, rh - 18)
wind_pm <- wind * 1.25

```

```

ffdi <- 2 * exp(-0.450 + 0.987 * log(pmax(drought, 0.1)) -
              0.0345 * rh_pm + 0.0338 * tmax + 0.0234 * wind_pm)
fri <- as.integer(cut(ffdi,
                     breaks = c(-Inf, 1, 2.5, 5, 8, 12, 17, 24, 35, 50, Inf),
                     labels = FALSE))

## 9. D'harawal seasons: onsets jittered +/- 10 days each year -----
season_tbl <- data.frame(
  Season      = c("Burran", "Marrai'gang", "Burrugin",
                  "Wiritjiribin", "Ngonungui", "Parra'dowee"),
  WeatherName = c("Gadalung Marool", "Bana'murrai'yung", "Tugarah Tuli",
                  "Tugarah Gunya'marri", "Murr'ai'yunggory", "Goray'murrai"),
  Plant       = c("Weetjellian (Lightwood) blooming",
                  "Lilly Pilly fruit ripening",
                  "Burringoa (Forest Red Gum) flowering",
                  "Marrai'uo (Gossamer Wattle) flowering",
                  "Miwa Gawaian (Waratah) blooming",
                  "Kai'arrewan (Coast Myall) blooming"),
  Animal      = c("Male kangaroos aggressive",
                  "Marrai'gang (quolls) calling for mates",
                  "Echidnas following mating lines",
                  "Lyrebirds building display mounds",
                  "Flying foxes gathering at dusk",
                  "Eels moving downriver to the sea"),
  stringsAsFactors = FALSE)

nominal <- c(91, 152, 213, 244, 305)      # nominal onsets, day-of-year
season_idx <- integer(n)
for (y in unique(yr)) {
  b <- sort(nominal + sample(-10:10, 5, replace = TRUE))
  k <- yr == y
  season_idx[k] <- findInterval(doy[k], b) + 1L
}

notes <- c(
  "Meat is not eaten during the hottest days; fruit and tubers gathered.",
  "Warm cloaks prepared; groups move toward the coast.",
  "Shellfish are not taken until the boo'kerrikin blooms.",
  "Gentle spring rains begin; lyrebird calls heard at dawn.",
  "Season of ceremony and gathering.",
  "Fish and eels are plentiful in the bays and estuaries.")

df <- data.frame(
  Date          = dates,
  StationID     = "SYD-CICS-01",
  Temperature   = round(temp, 1),
  MaxTemp       = round(tmax, 1),
  MinTemp       = round(temp - drange / 2, 1),
  Rainfall      = rain,
  Humidity      = round(rh, 1),
  WindSpeed     = round(wind, 1),
  SolarExposure = round(solar, 1),
  SoilMoisture  = round(sm, 1),

```

```

    FireRiskIndex      = fri,
    Season              = season_tbl$Season[season_idx],
    SeasonWeatherName  = season_tbl$WeatherName[season_idx],
    IndicatorPlant      = season_tbl$Plant[season_idx],
    IndicatorAnimal     = season_tbl$Animal[season_idx],
    CommunityNote       = notes[season_idx],
    stringsAsFactors   = FALSE)

df$Season <- factor(df$Season, levels = season_tbl$Season)
df
}

dataset_fingerprint <- function(df) {
  num <- unlist(df[vapply(df, is.numeric, logical(1))], use.names = FALSE)
  s <- as.integer(round(sum(num, na.rm = TRUE)))           %% 1000000L
  k <- as.integer(sum(utf8ToInt(paste0(as.character(df$Season),
                                     collapse = ""))))    %% 10000L
  d <- as.integer(which.max(df$MaxTemp) + 7 * which.max(df$Rainfall)) %% 1000L
  sprintf("WX-%06d-%04d-%03d", s, k, d)
}

## -----
## Generate YOUR dataset
## -----

weather_data <- generate_weather_data(params$studentID)

## Untouched copy of the data exactly as generated. Do not modify or delete it.
## The fingerprint must always be computed from this object, never from
## `weather_data`, because later parts of the assignment add columns to
## `weather_data` and that would change the fingerprint.
weather_data_orig <- weather_data

cat("Student ID :", params$studentID, "\n")

## Student ID : u3284342

cat("Seed      :", seed_from_id(params$studentID), "\n")

## Seed      : 12109058

cat("Rows x Cols :", nrow(weather_data), "x", ncol(weather_data), "\n")

## Rows x Cols : 731 x 16

cat("FINGERPRINT :", dataset_fingerprint(weather_data_orig), "\n")

## FINGERPRINT : WX-134679-6835-871

head(weather_data)

##      Date   StationID Temperature MaxTemp MinTemp Rainfall Humidity
## 1 2024-01-01 SYD-CICS-01      23.7    27.4    20.0      0.0     74.7
## 2 2024-01-02 SYD-CICS-01      24.3    27.1    21.5      0.0     71.1
## 3 2024-01-03 SYD-CICS-01      21.0    23.7    18.3      2.3     79.7
## 4 2024-01-04 SYD-CICS-01      20.8    22.3    19.2      2.7     83.9
## 5 2024-01-05 SYD-CICS-01      23.5    26.3    20.8      0.0     79.8
## 6 2024-01-06 SYD-CICS-01      26.4    30.4    22.4      0.0     82.4

```

```

##      WindSpeed SolarExposure SoilMoisture FireRiskIndex Season SeasonWeatherName
## 1         5.5         31.0         30.2           3 Burran   Gadalung Marool
## 2         8.8         22.3         27.8           3 Burran   Gadalung Marool
## 3         6.7         18.4         27.1           2 Burran   Gadalung Marool
## 4        31.0         10.0         26.9           3 Burran   Gadalung Marool
## 5        26.7         29.5         25.0           4 Burran   Gadalung Marool
## 6         9.5         31.2         22.2           3 Burran   Gadalung Marool
##
##      IndicatorPlant      IndicatorAnimal
## 1 Weetjellan (Lightwood) blooming Male kangaroos aggressive
## 2 Weetjellan (Lightwood) blooming Male kangaroos aggressive
## 3 Weetjellan (Lightwood) blooming Male kangaroos aggressive
## 4 Weetjellan (Lightwood) blooming Male kangaroos aggressive
## 5 Weetjellan (Lightwood) blooming Male kangaroos aggressive
## 6 Weetjellan (Lightwood) blooming Male kangaroos aggressive
##
##      CommunityNote
## 1 Meat is not eaten during the hottest days; fruit and tubers gathered.
## 2 Meat is not eaten during the hottest days; fruit and tubers gathered.
## 3 Meat is not eaten during the hottest days; fruit and tubers gathered.
## 4 Meat is not eaten during the hottest days; fruit and tubers gathered.
## 5 Meat is not eaten during the hottest days; fruit and tubers gathered.
## 6 Meat is not eaten during the hottest days; fruit and tubers gathered.

```

Part A: Data Understanding

Every claim in Part A must be supported by a number that appears in your own knitted output.

Task A.1: Describe the dataset in 120 words

This dataset contains 731 daily records (1 Jan 2024 to 31 Dec 2025) from one Sydney station, SYD-CICS-01. The measured variables are mean, maximum and minimum temperature, rainfall, humidity, wind speed, solar exposure, soil moisture and a 1–10 fire risk index. Mean temperature was 17.9 °C, the hottest maximum reached 37.4 °C, and 2512.4 mm of rain fell, with 433 dry days. Each day is also labelled with one of six D’harawal seasons, plus its weather name, indicator plant, indicator animal and a community note; Burran is the most frequent (172 days). The data supports comparing rainfall and soil moisture across the D’harawal seasons, and examining how soil moisture, humidity and temperature drive fire risk.

Task A.2: Summary statistics

```

## Number of days      : 731
## Date range          : 01 Jan 2024 31 Dec 2025
##
## Mean temperature (°C) : 17.9
## Highest max temp (°C) : 37.4
## Total rainfall (mm)   : 2512.4
## Dry days (0 mm)      : 433
## Days per season:
##
##      Burran  Marrai'gang  Burrugin Wiritjiribin  Ngoonungi  Parra'dowee
##      172      134      129      44      131      121

```

```
##
## --- Summary of the full data frame ---

##      Date      StationID      Temperature      MaxTemp
## Min.   :2024-01-01   Length   :731   Min.    : 6.2   Min.    : 7.90
## 1st Qu.:2024-07-01   N.unique  : 1   1st Qu.:14.1   1st Qu.:17.10
## Median :2024-12-31   N.blank  : 0   Median :18.2   Median :21.20
## Mean   :2024-12-31   Min.nchar: 11   Mean   :17.9   Mean   :20.92
## 3rd Qu.:2025-07-01   Max.nchar: 11   3rd Qu.:21.4   3rd Qu.:24.75
## Max.   :2025-12-31               Max.   :32.5   Max.   :37.40
##      MinTemp      Rainfall      Humidity      WindSpeed
## Min.    : 4.60   Min.    : 0.000   Min.    :33.80   Min.    : 1.50
## 1st Qu.:11.20   1st Qu.: 0.000   1st Qu.:63.15   1st Qu.: 9.65
## Median :15.10   Median : 0.000   Median :71.00   Median :14.40
## Mean    :14.88   Mean    : 3.437   Mean    :70.60   Mean    :15.07
## 3rd Qu.:18.50   3rd Qu.: 3.950   3rd Qu.:77.90   3rd Qu.:19.70
## Max.    :27.70   Max.    :67.300   Max.    :97.20   Max.    :41.70
## SolarExposure   SoilMoisture   FireRiskIndex      Season
## Min.    : 1.00   Min.    : 7.20   Min.    :1.000   Burran    :172
## 1st Qu.:10.25   1st Qu.:14.00   1st Qu.:3.000   Marrai'gang :134
## Median :14.80   Median :21.20   Median :3.000   Burrugin   :129
## Mean    :15.82   Mean    :22.21   Mean    :3.404   Wiritjiribin: 44
## 3rd Qu.:21.90   3rd Qu.:28.80   3rd Qu.:4.000   Ngoonungi  :131
## Max.    :31.70   Max.    :55.10   Max.    :7.000   Parra'dowee :121
## SeasonWeatherName IndicatorPlant IndicatorAnimal CommunityNote
## Length    :731   Length    :731   Length    :731   Length    :731
## N.unique   : 6   N.unique   : 6   N.unique   : 6   N.unique   : 6
## N.blank    : 0   N.blank    : 0   N.blank    : 0   N.blank    : 0
## Min.nchar  :12   Min.nchar  :26   Min.nchar  :25   Min.nchar  :33
## Max.nchar  :19   Max.nchar  :37   Max.nchar  :38   Max.nchar  :69
##

##      Mean      SD SD_to_Mean
## Rainfall      3.437  6.780      1.973
## WindSpeed     15.067  7.238      0.480
## SolarExposure 15.816  7.517      0.475
## SoilMoisture  22.212 10.284      0.463
## FireRiskIndex 3.404  1.154      0.339
## MinTemp       14.877  4.516      0.304
## Temperature   17.902  4.835      0.270
## MaxTemp       20.921  5.262      0.251
## Humidity      70.604  9.921      0.141
##

## Largest SD/mean ratio: Rainfall ( 1.973 )

##
## Class of Season      : factor
## Class of Temperature: numeric

##      Burran Marrai'gang Burrugin Wiritjiribin Ngoonungi Parra'dowee
##      172      134      129      44      131      121

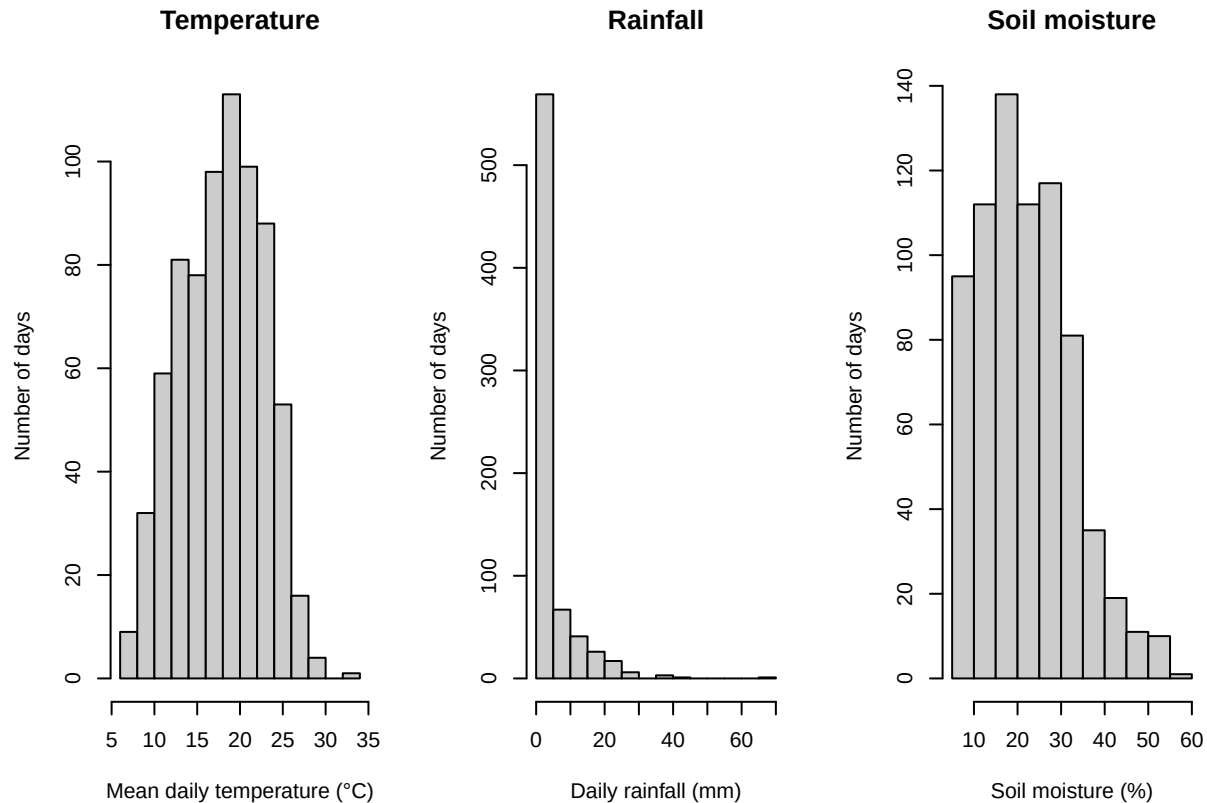
##      Min. 1st Qu. Median Mean 3rd Qu. Max.
##      6.2  14.1  18.2  17.9  21.4  32.5
```

a. The numeric variable with the largest ratio of standard deviation to mean is Rainfall (1.973, an SD

of 6.78 mm against a mean of 3.437 mm), which tells me that daily rainfall is extremely variable and right-skewed, with a median of 0 mm and a maximum of 67.3 mm, so a few heavy-rain days supply most of the total and the mean does not describe a typical day.

- b. `summary()` treats `Season` differently from `Temperature` because `Season` is a factor while `Temperature` is numeric, so it reports the count of days in each level (e.g. Burran 172, Wiritjiribin 44) instead of quartiles and a mean (17.9 °C for Temperature). To get a mean for it I would have to convert it to numeric with `as.numeric(weather_data$Season)`, giving level codes 1–6, although that mean would be meaningless because the seasons are categories, not quantities.

Task A.3: Histograms



Task A.4: Read your own plots (max 150 words total)

```
## Dry days           : 433
## First bar          : 0 to 5 mm, height 569
## Difference          : 136

##           SoilMoisture Rainfall
## Burran      17.71      3.73
## Marrai'gang 24.67      3.78
## Burrugin    32.57      3.65
## Wiritjiribin 33.16      3.88
## Ngoonungi   19.91      2.36
## Parra'dowee 13.37      3.42

##
## Highest SoilMoisture: Wiritjiribin
```

Highest Rainfall : Wiritjiribin

- The shape of my Temperature histogram is broad and slightly left-skewed (mean 17.9 °C, median 18.2 °C), with a steep drop above 25 °C and a thin tail to 32.5 °C, and the data were generated that way because the annual cosine climatology spreads days across a wide range, rain days are cooled by 1.6 °C, and rare summer heatwave spikes form the thin right tail.
 - My number of dry days is 433. The height of the first bar of the histogram is 569. The difference of 136 days is made up of wet days with 0.2 to 5 mm of rain.
 - The season with the highest mean SoilMoisture is Wiritjiribin (value 33.16 %). It is the same season with the highest mean Rainfall (3.88 mm/day), because its rain adds to water already stored through cool, low-evaporation Burrugin (32.57 %).
-

Part B: Vectors, Matrices and Arrays

Throughout Part B, “month” means a calendar month within a specific year (2024-01 ... 2025-12), giving 24 groups.

B.1 — avg_temp_month

```
## 2024-01 2024-02 2024-03 2024-04 2024-05 2024-06 2024-07 2024-08 2024-09 2024-10
## 22.78 21.73 21.12 18.28 15.49 14.44 12.34 11.81 15.60 19.91
## 2024-11 2024-12 2025-01 2025-02 2025-03 2025-04 2025-05 2025-06 2025-07 2025-08
## 20.12 22.67 23.67 23.32 20.48 16.78 15.78 13.08 11.21 10.86
## 2025-09 2025-10 2025-11 2025-12
## 15.62 19.46 20.53 23.04

##
## Maximum: 23.67 °C in 2025-01
```

The maximum of avg_temp_month is 23.67 °C in January 2025 2025-01 .

B.2 — avg_rain_month, avg_hum_month, avg_soil_month

```
## 2024-01 2024-02 2024-03 2024-04 2024-05 2024-06 2024-07 2024-08 2024-09 2024-10
## 2.60 6.04 4.74 3.84 5.00 2.35 7.04 3.93 1.82 2.74
## 2024-11 2024-12 2025-01 2025-02 2025-03 2025-04 2025-05 2025-06 2025-07 2025-08
## 0.53 5.79 3.15 2.43 3.55 2.85 3.51 2.62 3.15 2.08
## 2025-09 2025-10 2025-11 2025-12
## 3.70 2.55 1.99 4.32

## 2024-01 2024-02 2024-03 2024-04 2024-05 2024-06 2024-07 2024-08 2024-09 2024-10
## 73.15 76.51 71.32 71.84 70.74 66.11 70.39 70.87 66.26 64.67
## 2024-11 2024-12 2025-01 2025-02 2025-03 2025-04 2025-05 2025-06 2025-07 2025-08
## 69.32 73.34 69.45 71.01 74.40 74.17 70.44 69.38 71.82 75.21
## 2025-09 2025-10 2025-11 2025-12
## 67.31 66.82 68.38 71.62

## 2024-01 2024-02 2024-03 2024-04 2024-05 2024-06 2024-07 2024-08 2024-09 2024-10
## 17.09 23.72 24.89 22.68 33.25 34.64 45.77 37.62 27.86 14.07
## 2024-11 2024-12 2025-01 2025-02 2025-03 2025-04 2025-05 2025-06 2025-07 2025-08
## 8.78 19.81 12.50 11.65 17.50 16.25 24.65 23.18 29.82 24.00
## 2025-09 2025-10 2025-11 2025-12
## 25.29 12.35 11.12 13.43
```

B.3 — Matrix M_month (24 x 4)

```
## [1] 24 4
##      Temperature_C Humidity_pct WindSpeed_kmh SoilMoisture_pct
## 2024-01          22.78         73.15         14.55          17.09
## 2024-02          21.73         76.51         14.47          23.72
## 2024-03          21.12         71.32         15.21          24.89
## 2024-04          18.28         71.84         14.39          22.68
## 2024-05          15.49         70.74         14.45          33.25
## 2024-06          14.44         66.11         14.37          34.64
## 2024-07          12.34         70.39         14.70          45.77
## 2024-08          11.81         70.87         14.15          37.62
## 2024-09          15.60         66.26         17.63          27.86
## 2024-10          19.91         64.67         14.60          14.07
## 2024-11          20.12         69.32         16.19           8.78
## 2024-12          22.67         73.34         16.20          19.81
## 2025-01          23.67         69.45         15.30          12.50
## 2025-02          23.32         71.01         15.25          11.65
## 2025-03          20.48         74.40         13.80          17.50
## 2025-04          16.78         74.17         13.78          16.25
## 2025-05          15.78         70.44         15.84          24.65
## 2025-06          13.08         69.38         14.47          23.18
## 2025-07          11.21         71.82         15.75          29.82
## 2025-08          10.86         75.21         13.30          24.00
## 2025-09          15.62         67.31         16.49          25.29
## 2025-10          19.46         66.82         15.75          12.35
## 2025-11          20.53         68.38         15.85          11.12
## 2025-12          23.04         71.62         15.16          13.43
```

B.4 — Matrix M_season

```
##      Temperature Rainfall Humidity WindSpeed SolarExposure
## Burran          22.3      3.7      72.7      14.9          19.8
## Marrai'gang     16.9      3.8      71.2      14.4          10.9
## Burrugin        12.3      3.7      70.6      14.7           9.2
## Wiritjiribin    11.5      3.9      72.3      13.8          12.0
## Ngoonungi       17.4      2.4      66.3      15.9          18.1
## Parra'dowee     21.6      3.4      70.9      15.9          21.6
```

B.5 — Array A_month (year x month x variable)

```
## $Year
## [1] "2024" "2025"
##
## $Month
## [1] "Jan" "Feb" "Mar" "Apr" "May" "Jun" "Jul" "Aug" "Sep" "Oct" "Nov" "Dec"
##
## $Variable
## [1] "Temperature" "Rainfall" "Humidity" "WindSpeed"
## [5] "SolarExposure" "FireRiskIndex"
##
## 2024 2025
## 8.54 9.16
```

The July SolarExposure values are 8.54 in 2024 and 9.16 in 2025.

B.6 — `t(M_month) %*% avg_temp_month`

```
## [1] 4 1
##
##           [,1]
## Temperature_C      8114.20
## Humidity_pct       30398.08
## WindSpeed_kmh       6499.86
## SoilMoisture_pct    8868.60
```

The units of the first element are ..., and the result is not a meaningful summary of my data because ...

B.7 — `above_below()` and the `TempFlag` column

```
## Total 'a' days: 365 of 731
## 2024-01 2024-02 2024-03 2024-04 2024-05 2024-06 2024-07 2024-08 2024-09 2024-10
##    0.645    0.552    0.484    0.500    0.516    0.400    0.452    0.484    0.500    0.484
## 2024-11 2024-12 2025-01 2025-02 2025-03 2025-04 2025-05 2025-06 2025-07 2025-08
##    0.567    0.516    0.516    0.536    0.419    0.500    0.452    0.567    0.484    0.452
## 2025-09 2025-10 2025-11 2025-12
##    0.533    0.484    0.500    0.452
##
## Highest proportion: 2024-01 ( 0.645 )
```

Total number of “a” days: ... Month with the highest proportion of “a” days: ... (proportion ...)

The total is not exactly half of 731 because 2024 having 366 days due to it being a leap year

B.8 — Year-on-year comparison

```
##   January  February    March    April      May      June      July      August
##     0.89     1.58    -0.64    -1.50     0.29    -1.36    -1.13    -0.95
## September  October  November  December
##     0.02    -0.44     0.40     0.37
##
## Largest warming: February ( 1.58 °C)
```

```
## Largest cooling: April ( -1.5 °C)
```

Largest warming: February (1.58 °C). Largest cooling: April (-1.5 °C).

Part C: Looping and Conditional Statements

Each chunk below must use an explicit loop or conditional as described, and must contain **one comment line in your own words saying *why* you structured the loop the way you did** — not what the line does.

C.1 — Mean SoilMoisture for each of the 24 months

```
## 2024-01 2024-02 2024-03 2024-04 2024-05 2024-06 2024-07 2024-08 2024-09 2024-10
##   17.09   23.72   24.89   22.68   33.25   34.64   45.77   37.62   27.86   14.07
## 2024-11 2024-12 2025-01 2025-02 2025-03 2025-04 2025-05 2025-06 2025-07 2025-08
##    8.78   19.81   12.50   11.65   17.50   16.25   24.65   23.18   29.82   24.00
## 2025-09 2025-10 2025-11 2025-12
```

25.29 12.35 11.12 13.43

##

Highest: 2024-07 (45.77 %)

Lowest : 2024-11 (8.78 %)

Highest: 45.77 % (month 2024-07). Lowest: 8.78 % (month 2024-11).

C.2 — Days with Temperature above 26 °C

Days above 26 °C: 21

Percentage : 2.87 %

Count of Days with a Temperature above 26 °C 21 days , which is 2.87% of the record.

C.3 — Mean Humidity on days below 15 °C

Days below 15 °C : 214

Mean humidity (< 15 °C): 72.82 %

Mean humidity (others) : 69.69 %

Mean humidity below 15 °C: ... ; all other days: The difference goes in that direction because ...

C.4 — FireRiskIndex above 4 in Burran and Burrugin

Burran days with FireRiskIndex > 4: 39 of 172

Burrugin days with FireRiskIndex > 4: 3 of 129

Burran: ... days. Burrugin: ... days. Each count is / is not consistent with its D'harawal weather name because ...

C.5 — Total Rainfall matrix (24 months x 6 seasons)

##	Season						
##	Month	Burran	Marrair'gang	Burrugin	Wiritjiribin	Ngoonungi	Parra'dowee
##	2024-01	80.5	0.0	0.0	0.0	0.0	0.0
##	2024-02	175.2	0.0	0.0	0.0	0.0	0.0
##	2024-03	123.1	23.8	0.0	0.0	0.0	0.0
##	2024-04	0.0	115.3	0.0	0.0	0.0	0.0
##	2024-05	0.0	154.9	0.0	0.0	0.0	0.0
##	2024-06	0.0	5.4	65.0	0.0	0.0	0.0
##	2024-07	0.0	0.0	218.2	0.0	0.0	0.0
##	2024-08	0.0	0.0	0.0	121.8	0.0	0.0
##	2024-09	0.0	0.0	0.0	0.0	54.5	0.0
##	2024-10	0.0	0.0	0.0	0.0	54.1	30.9
##	2024-11	0.0	0.0	0.0	0.0	0.0	15.9
##	2024-12	0.0	0.0	0.0	0.0	0.0	179.6
##	2025-01	97.7	0.0	0.0	0.0	0.0	0.0
##	2025-02	68.0	0.0	0.0	0.0	0.0	0.0
##	2025-03	97.4	12.5	0.0	0.0	0.0	0.0
##	2025-04	0.0	85.6	0.0	0.0	0.0	0.0
##	2025-05	0.0	108.8	0.0	0.0	0.0	0.0
##	2025-06	0.0	0.0	78.5	0.0	0.0	0.0
##	2025-07	0.0	0.0	97.7	0.0	0.0	0.0

```
## 2025-08 0.0 0.0 11.9 48.7 3.9 0.0
## 2025-09 0.0 0.0 0.0 0.0 110.9 0.0
## 2025-10 0.0 0.0 0.0 0.0 79.0 0.0
## 2025-11 0.0 0.0 0.0 0.0 6.2 53.6
## 2025-12 0.0 0.0 0.0 0.0 0.0 133.8
```

```
##
```

```
## Cells exactly zero: 113 of 144
```

```
## Zero because season absent : 111
```

```
## Zero despite season present : 2
```

```
##      Date      From      To
## 1 2024-03-28      Burran Marrai'gang
## 2 2024-06-06 Marrai'gang      Burrugin
## 3 2024-08-02      Burrugin Wiritjiribin
## 4 2024-09-01 Wiritjiribin      Ngoonungi
## 5 2024-10-30      Ngoonungi Parra'dowee
## 6 2025-01-01 Parra'dowee      Burran
## 7 2025-03-27      Burran Marrai'gang
## 8 2025-05-30 Marrai'gang      Burrugin
## 9 2025-08-10      Burrugin Wiritjiribin
## 10 2025-08-24 Wiritjiribin      Ngoonungi
## 11 2025-11-04      Ngoonungi Parra'dowee
```

a. Cells that are exactly zero: ... out of 144.

b. So many cells are zero because The pattern of the non-zero cells tells me that ...

c. Month containing two seasons: ... ; transition date: ...

C.6 — Longest dry spell

```
## Longest dry spell : 15 days
```

```
## Start date      : 2025-05-28
```

```
## End date        : 2025-06-11
```

```
## SoilMoisture start: 30.6 %
```

```
## SoilMoisture end : 18.1 %
```

Length: ... days. Start: End: SoilMoisture on the first day: ... ; on the last day: The change goes in that direction because ...

Part D: Reasoning, Prediction and Verification

D.1 — Explain your own code (max 150 words)

I chose Part C task 5. First, `months` is a character vector of the 24 month labels and `season_levels` is a character vector of the 6 season names. `rain_mat` is pre-allocated as a 24×6 numeric matrix of zeros, labelled with those names. The outer loop takes month `i` and the inner loop season `j`; for each pair, `rows` is a logical vector of length 731 that is TRUE where a day falls in both, and the rainfall on those days is summed into cell `[i, j]`. After both loops, `rain_mat` is a 24×6 numeric matrix of rainfall totals in mm. The line `rows <- month_id == months[i] & ...` would break with a different number of rows, because `month_id` was

built once in B.1 with 731 elements; if `weather_data` changed length, the two vectors would no longer line up, giving recycled, misaligned or NA totals.

D.2 — Predict, then check

Write your predictions in the chunk below **before running anything**.

PREDICTION (write BEFORE running):

Q1. Roughly how many days in your record have Rainfall above 20 mm?

My prediction: _____

**Q2. Will the median of Rainfall be greater or less than the mean?
By roughly how much?**

My prediction: _____

Q3. What proportion of the 731 days in your record are dry (Rainfall == 0)?

My prediction: _____

Now run the three lines below and reconcile.

```
## [1] 28
##      mean  median
## 3.436936 0.000000
## [1] 0.5923393
```

Reconciliation (max 80 words):

Q1 actual ... — my prediction was wrong / right because ... Q2 actual ... — ... Q3 actual ... — ...

D.3 — Test four claims against your data

```
##           Jan  Feb  Mar  Apr  May  Jun  Jul  Aug  Sep  Oct  Nov
## TotalRain_mm 178.2 243.2 256.8 200.9 263.7 148.9 315.9 186.3 165.4 164.0 75.7
## MeanSoil_pct 14.8  17.8  21.2  19.5  28.9  28.9  37.8  30.8  26.6  13.2 10.0
##           Dec
## TotalRain_mm 313.4
## MeanSoil_pct 16.6

## Wettest month: Jul | Highest soil moisture: Jul | Claim 1: TRUE

## 2024 2025
## 18.014 17.788

## Claim 2: FALSE

## Days above 25 °C: 40 | Claim 3: TRUE
```

```
## Longest dry spell: 15 days | Claim 4: FALSE
```

Claim 1: numbers ... -> TRUE / FALSE Claim 2: numbers ... -> TRUE / FALSE Claim 3: numbers ...
-> TRUE / FALSE Claim 4: numbers ... -> TRUE / FALSE

Claim 1 can be false even though rainfall is the only source of soil moisture because ... (two sentences)

Claims 2 and 3 came out differently for different students because ... (one sentence)

D.4 — `classify_day()` with thresholds from your own distribution

```
## 20% 80%
```

```
## 13.2 22.6
```

```
##
```

```
## Cool Mild Hot
```

```
## 146 443 142
```

Quantile cut-points used: ... and ... (from `quantile()`). Counts: Hot ... , Mild ... , Cool

A quantile-based threshold is better than a fixed one on another site's data because ...

D.5 — Debugging: code that runs but is wrong

Both lines use `weather_data_orig`, the unmodified data frame from the setup chunk.

```
## [1] 817.8813
```

```
## [1] 65.65905
```

Now compute the correct versions, and the number of days each version of line (ii) selects.

```
## (i) wrong: 817.88 | correct: 17.9 | divided by 16 instead of 731
```

```
## (ii) wrong (|): 65.66 % from 442 days
```

```
## (ii) right (&): 61.26 % from 31 days
```

Line (i): wrong value ... ; correct value What R actually did: ... , and the mistake is silent because ...

Line (ii): wrong value ... (selects ... days); correct value ... (selects ... days). What R actually did: ... , and the mistake is silent because ...

D.6 — Interpretation grounded in your own results

```
##           Season MeanTemp Days
## Burran      Burran    22.25  172
## Parra'dowee Parra'dowee 21.59  121
## Ngoonungi   Ngoonungi  17.45  131
## Marrai'gang Marrai'gang  16.94  134
## Burrugin    Burrugin   12.31  129
## Wiritjiribin Wiritjiribin 11.45   44
```

```
##           MeanRain_mm_per_day PropRainDays TotalRain_mm
## Burran      3.732      0.442      641.9
## Marrai'gang 3.778      0.485      506.3
## Burrugin    3.653      0.395      471.3
## Wiritjiribin 3.875      0.386      170.5
## Ngoonungi   2.356      0.298      308.6
## Parra'dowee 3.420      0.413      413.8
```

- a. Seasons ranked by mean Temperature, with the number of days each covers:
 1. ... (... °C, ... days)
 2. ...
 3. ...
 4. ...
 5. ...
 6. ...
- b. “Hot”: “Dry” tested as a rate: My dataset does / does not support both halves. A seasonal total would mislead here because ...
- c. My dataset cannot be used as evidence for or against the Parra’dowee note because ... The data that would be needed instead are ... (two to three sentences)

D.7 — Write your own classifier

Median wet-day rainfall: 5.8 mm | 90th percentile wind: 24.7 km/h

##

Dry Stormy Wet

433 21 277

Date Rainfall WindSpeed MaxTemp

196 2024-07-14 35.5 29.3 11.2

Date Rainfall WindSpeed WeatherClass

71 2024-03-11 67.3 6.5 Wet

Date Rainfall WindSpeed WeatherClass

701 2025-12-01 0 39.2 Dry

Thresholds and the statistic justifying each: - Wet: ... because ... - Stormy: ... because ...

Counts: Wet ... , Dry ... , Stormy

Stormiest date: ... — Rainfall ... mm, WindSpeed ... km/h, MaxTemp ... °C.

One day I disagree with: ... , because I would change ...

Verification and Reproducibility

Results Summary Table

Fill in this table **manually** from your own output above. Rows 1-11 are checked automatically against your student ID; row 12 is marked by inspection. Leave no cell blank; write **NA** and explain if you could not compute one.

#	Quantity	Your value
1	Dataset fingerprint (from the setup chunk)	WX-134679-6835-871
2	Mean Temperature over the whole record (2 dp)	17.90
3	Total Rainfall over the whole record (1 dp)	2512.4
4	Number of days with Rainfall == 0	433
5	Date of the highest MaxTemp , and that value	
6	Season with the highest mean Temperature , and that mean (2 dp)	
7	Number of days with FireRiskIndex > 4 in Burran	
8	Length of the longest dry spell (Part C task 6)	

#	Quantity	Your value
9	Start date of the longest dry spell (earliest run, if tied)	
10	<code>cor(SoilMoisture, FireRiskIndex)</code> (2 dp)	
11	Number of "a" days from Part B task 7	365
12	Your two <code>classify_day()</code> quantile cut-points (D.4)	

Reproducibility appendix

Leave this chunk unchanged as the **last chunk** of your submission.

```
cat("Student ID :", params$studentID, "\n")

## Student ID : u3284342

cat("Fingerprint :", dataset_fingerprint(weather_data_orig), "\n")

## Fingerprint : WX-134679-6835-871

cat("Knitted at :", format(Sys.time(), "%Y-%m-%d %H:%M:%S %Z"), "\n")

## Knitted at : 2026-09-28 19:11:59 AEST

cat("Chunks run :", length(knitr::all_labels()), "\n\n")

## Chunks run : 29

print(sessionInfo())$R.version$version.string

## [1] "R version 4.6.1 (2026-06-24)"
```

Declaration

I confirm that the dataset analysed in this submission was generated from my own student ID, that the values reported in the Results Summary Table are the values produced by the code in this document, and that all explanations and interpretations are my own work.

Signed: Ruhan Shafi Date: 28th of September, 2026

Overall Conclusion

Your overall reflection about this assignment goes here ...